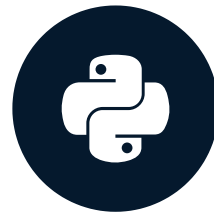


Introduction to relational databases

INTRODUCTION TO IMPORTING DATA IN PYTHON



Hugo Bowne-Anderson
Data Scientist at DataCamp

What is a relational database?

- Based on relational model of data
- First described by Edgar “Ted” Codd

Example: Northwind database

- Orders table

OrderID	CustomerID	EmployeeID	OrderDate	RequiredDate	ShippedDate	ShipVia	Freight	ShipName	ShipAddress
10248	VINET	5	7/4/1996 12:00:00 AM	8/1/1996 12:00:00 AM	7/16/1996 12:00:00 AM	3	32.38	Vins et alcools Chevalier	59 rue de l'Abbaye
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- Customers table

CustomerID	CompanyName	ContactName	ContactTitle	Address	City	Region	PostalCode	Country
ALFKI	Alfreds Futterkiste	Maria Anders	Sales Representative	Obere Str. 57	Berlin	None	12209	Germany
AROUT	Around the Horn	Thomas Hardy	Sales Representative	120 Hanover Sq.	London	None	WA1 1DP	UK
BLAUS	Blauer See Delikatessen	Hanna Moos	Sales Representative	Forsterstr. 57	Mannheim	None	68306	Germany
BONAP	Bon app'	Laurence Lebihan	Owner	12, rue des Bouchers	Marseille	None	13008	France

- Employees table

EmployeeID	LastName	FirstName	Title	TitleOfCourtesy	BirthDate	HireDate	Address	City	Region
1	Davolio	Nancy	Sales Representative	Ms.	12/8/1948 12:00:00 AM	5/1/1992 12:00:00 AM	507 - 20th Ave. E.\nApt. 2A	Seattle	WA
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Relational model

- Widely adopted
- Codd's 12 Rules/Commandments
 - Consists of 13 rules (zero-indexed!)
 - Describes what a Relational Database Management System should adhere to to be considered relational

Relational Database Management Systems

- PostgreSQL



Relational Database Management Systems

- PostgreSQL
- MySQL



Relational Database Management Systems

- PostgreSQL
- MySQL
- SQLite



Relational Database Management Systems

- PostgreSQL
- MySQL
- SQLite
- SQL = Structured Query Language

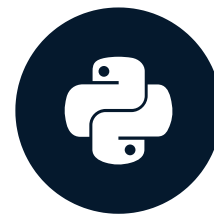


Let's practice!

INTRODUCTION TO IMPORTING DATA IN PYTHON

Creating a database engine in Python

INTRODUCTION TO IMPORTING DATA IN PYTHON



Hugo Bowne-Anderson
Data Scientist at DataCamp

Creating a database engine

- SQLite database
 - Fast and simple
- SQLAlchemy
 - Works with many Relational Database Management Systems

```
from sqlalchemy import create_engine  
engine = create_engine('sqlite:///Northwind.sqlite')
```

Getting table names

```
from sqlalchemy import create_engine  
engine = create_engine('sqlite:///Northwind.sqlite')
```

```
table_names = engine.table_names()  
print(table_names)
```

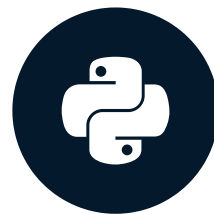
```
['Categories', 'Customers', 'EmployeeTerritories',  
'Employees', 'Order Details', 'Orders', 'Products',  
'Region', 'Shippers', 'Suppliers', 'Territories']
```

Let's practice!

INTRODUCTION TO IMPORTING DATA IN PYTHON

Querying relational databases in Python

INTRODUCTION TO IMPORTING DATA IN PYTHON



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Basic SQL query

```
SELECT * FROM Table_Name
```

- Returns all columns of all rows of the table
- Example:

```
SELECT * FROM Orders
```

- We'll use SQLAlchemy and pandas

Workflow of SQL querying

- Import packages and functions
- Create the database engine
- Connect to the engine
- Query the database
- Save query results to a DataFrame
- Close the connection

Your first SQL query

```
from sqlalchemy import create_engine
import pandas as pd
engine = create_engine('sqlite:///Northwind.sqlite')
con = engine.connect()
rs = con.execute("SELECT * FROM Orders")
df = pd.DataFrame(rs.fetchall())
con.close()
```

Printing your query results

```
print(df.head())
```

	0	1	2	3	4
0	10248	VINET	5	7/4/1996 12:00:00 AM	8/1/1996 12:00:00 AM
1	10251	VICTE	3	7/8/1996 12:00:00 AM	8/5/1996 12:00:00 AM
2	10254	CHOPS	5	7/11/1996 12:00:00 AM	8/8/1996 12:00:00 AM
3	10256	WELLI	3	7/15/1996 12:00:00 AM	8/12/1996 12:00:00 AM
4	10258	ERNSH	1	7/17/1996 12:00:00 AM	8/14/1996 12:00:00 AM

Set the DataFrame column names

```
from sqlalchemy import create_engine
import pandas as pd
engine = create_engine('sqlite:///Northwind.sqlite')
con = engine.connect()
rs = con.execute("SELECT * FROM Orders")
df = pd.DataFrame(rs.fetchall())
df.columns = rs.keys()
con.close()
```

Set the data frame column names

```
print(df.head())
```

	OrderID	CustomerID	EmployeeID	OrderDate
0	10248	VINET	5	7/4/1996 12:00:00 AM
1	10251	VICTE	3	7/8/1996 12:00:00 AM
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Using the context manager

```
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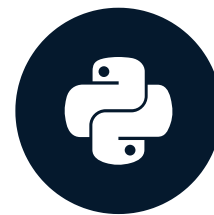
```
with engine.connect() as con:
    rs = con.execute("SELECT OrderID, OrderDate, ShipName FROM Orders")
    df = pd.DataFrame(rs.fetchmany(size=5))
    df.columns = rs.keys()
```


Let's practice!

INTRODUCTION TO IMPORTING DATA IN PYTHON

Querying relational databases directly with pandas

INTRODUCTION TO IMPORTING DATA IN PYTHON



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Data Scientist at DataCamp

The pandas way to query

```
from sqlalchemy import create_engine
import pandas as pd
engine = create_engine('sqlite:///Northwind.sqlite')
with engine.connect() as con:
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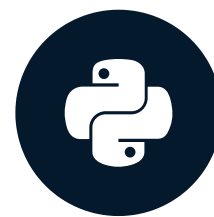
df = pd.read_sql_query("SELECT * FROM Orders", engine)
```

Let's practice!

INTRODUCTION TO IMPORTING DATA IN PYTHON

Advanced querying: exploiting table relationships

INTRODUCTION TO IMPORTING DATA IN PYTHON



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Tables are linked

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1	Davolio	Nancy	Sales Representative	Ms.	12/8/1948 12:00:00 AM	5/1/1992 12:00:00 AM	507 - 20th Ave. E.\nApt. 2A	Seattle	WA
2	Fuller	Andrew	Vice President, Sales	Dr.	2/19/1952 12:00:00 AM	8/14/1992 12:00:00 AM	908 W. Capital Way	Tacoma	WA
3	Leverling	Janet	Sales Representative	Ms.	8/30/1963 12:00:00 AM	4/1/1992 12:00:00 AM	722 Moss Bay Blvd.	Kirkland	WA

JOINing tables

- Orders table

OrderID	CustomerID	EmployeeID	OrderDate	RequiredDate	ShippedDate	ShipVia	Freight	ShipName	ShipAddress
10248	VINET	5	7/4/1996 12:00:00 AM	8/1/1996 12:00:00 AM	7/16/1996 12:00:00 AM	3	32.38	Vins et alcools Chevalier	59 rue de l'Abbaye
10251	VICTE	3	7/8/1996 12:00:00 AM	8/5/1996 12:00:00 AM	7/15/1996 12:00:00 AM	1	41.34	Victuailles en stock	2, rue du Commerce
10254	CHOPS	5	7/11/1996 12:00:00 AM	8/8/1996 12:00:00 AM	7/23/1996 12:00:00 AM	2	22.98	Chop-suey Chinese	Hauptstr. 31

- Customers table

CustomerID	CompanyName	ContactName	ContactTitle	Address	City	Region	PostalCode	Country
ALFKI	Alfreds Futterkiste	Maria Anders	Sales Representative	Obere Str. 57	Berlin	None	12209	Germany
AROUT	Around the Horn	Thomas Hardy	Sales Representative	120 Hanover Sq.	London	None	WA1 1DP	UK
BLAUS	Blauer See Delikatessen	Hanna Moos	Sales Representative	Forsterstr. 57	Mannheim	None	68306	Germany
BONAP	Bon app'	Laurence Lebihan	Owner	12, rue des Bouchers	Marseille	None	13008	France

JOINing tables

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INNER JOIN in Python (pandas)

```
from sqlalchemy import create_engine
import pandas as pd
engine = create_engine('sqlite:///Northwind.sqlite')
df = pd.read_sql_query("SELECT OrderID, CompanyName FROM Orders
INNER JOIN Customers on Orders.CustomerID = Customers.CustomerID", engine)
print(df.head())
```

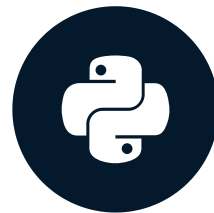
	OrderID	CompanyName
0	10248	Vins et alcools Chevalier
1	10251	Victuailles en stock
2	10254	Chop-suey Chinese
3	10256	Wellington Importadora
4	10258	Ernst Handel

Let's practice!

INTRODUCTION TO IMPORTING DATA IN PYTHON

Final Thoughts

INTRODUCTION TO IMPORTING DATA IN PYTHON



Hugo Bowne-Anderson
Data Scientist at DataCamp

What you've learned:

- Relational databases
- Queries
 - SELECT
 - WHERE
 - JOIN

Next course:

- Scrape data from the web
- Interact with APIs

Let's practice!

INTRODUCTION TO IMPORTING DATA IN PYTHON